Water Quality Archive: Data Quality Audit

What this is

A reproducible data-quality audit built against the Environment Agency's public Water Quality Archive API. A portable work sample demonstrating cross-reference, root-cause, and caveat workflows end to end.

Open data, open-source code (MIT licence). No proprietary or client material is involved; every finding is traceable to a public API endpoint and every check reproduces from the code in the repository.

It is not a claim that the audit surfaces a new problem for the Agency. Formatting drift in a long-lived reference codelist is expected signal, not an indictment.

Data pulled from the Water Quality Archive

```
8,070   determinand codelist records
  113   sampling-point-type codelist records
  200   sampling-point sample (JSON-LD)
```

Headline findings

```
1,514   determinand rows share a prefLabel with another row
        (e.g. 'Benzene' 6x, 'Dichloromethane' 6x, 'Enterovirus' 6x)
  502   near-duplicate label pairs at >= 90% fuzzy match
        (first 1,500 records scanned; see caveat, page 3)
  301   prefLabel formatting inconsistencies
        (whitespace, double spaces, all-caps, non-ASCII)
    0   referential-integrity orphans in the 200-point sample
        (sampling-point -> samplingPointType resolves cleanly)
    0   determinand records missing altLabel
```

How to read this document

Page 2: every finding in plain English, with a severity-coloured chart.
Page 3: methodology, the honest near-duplicate caveat, and attribution.

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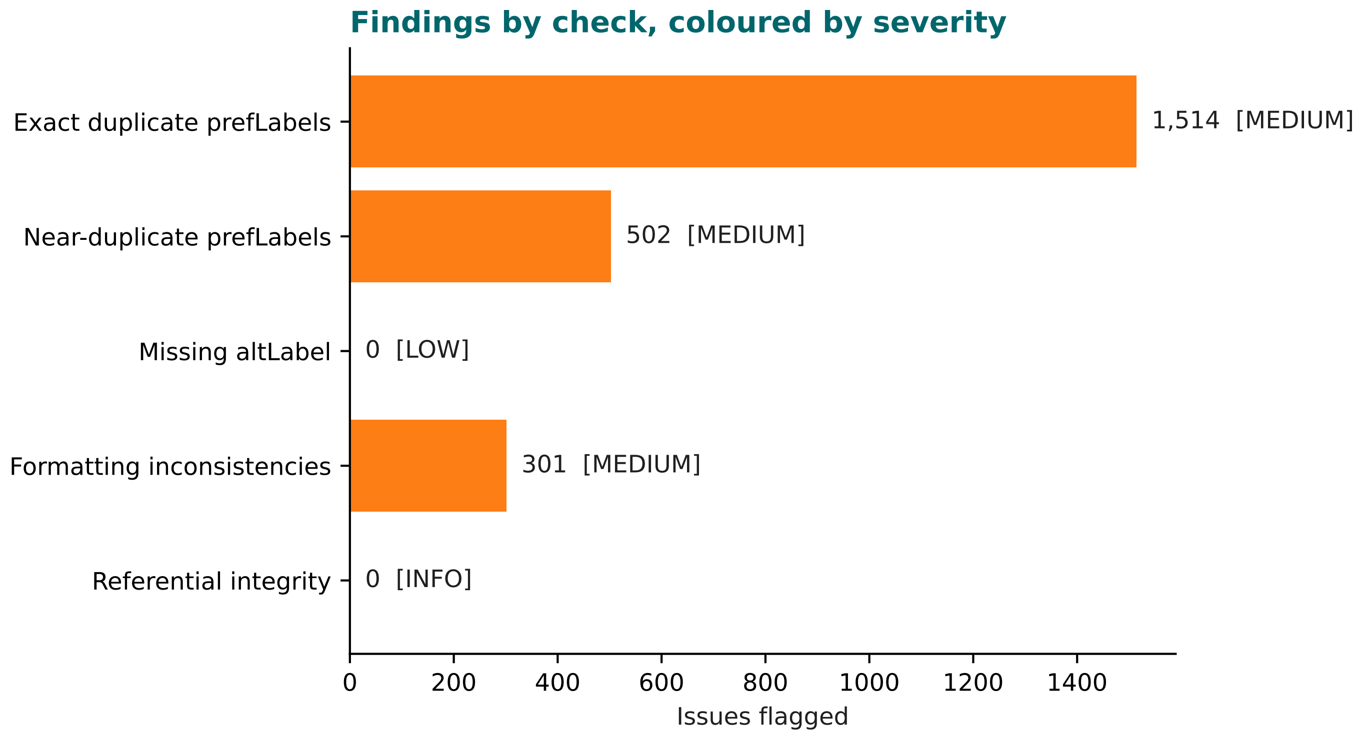

What each finding means

MEDIUM : Exact duplicate prefLabels in determinand codelist

1,122 distinct prefLabels appear on more than one row across 8,070 determinands. Examples: 'Benzene' 6x, 'Dichloromethane' 6x, 'Enterovirus' 6x. Consolidation candidates for codelist governance.

MEDIUM : Near-duplicate prefLabels (rapidfuzz >= 90%)

502 label pairs score >= 90% similarity by Levenshtein ratio within the first 1,500 records. Many are chemically distinct PFAS variants that differ by one letter. See the caveat on page 3.

LOW : Missing altLabel on determinand records

altLabel is populated on every determinand record in the snapshot. No gaps to surface.

MEDIUM : prefLabel formatting inconsistencies

301 formatting issues across the codelist: 38 leading or trailing whitespace, 12 double spaces, 249 all-caps runs, 2 non-ASCII. Drift from decades of collection methodology: expected signal, not data-entry error.

INFO : Sampling-point -> samplingPointType referential integrity

All samplingPointType codes cited by the 200-point sample resolve cleanly against the codelist. No orphans.

Methodology

- Every check is reproducible from code. API responses are cached in data/ so the audit re-runs offline against the same snapshot.
- The near-duplicate scan is capped at the first 1,500 determinand records (of 8,070) so the demo runs in under a minute. A production run would use first-character blocking to scan the full codelist linearly.
- rapidfuzz.ratio is Levenshtein-based: suggestive of duplication, not conclusive. Every pair flagged would be reviewed manually against source documentation before any codelist merge.

Honest caveat on the near-duplicate finding

Most of the 502 pairs flagged at >= 90% are not duplicates. They are chemically distinct PFAS variants whose names differ by one letter. Ethyl vs methyl substitution changes the molecule but not the label:

2-(N-Ethylperfluorooctanesulfonamido)acetic acid
2-(N-Methylperfluorooctanesulfonamido)acetic acid

rapidfuzz scores those at 99%. A subject-matter expert would correctly keep them separate. This is a deliberate demonstration of where automated DQ rules need manual SME review. It is not a claim that the codelist is 502 items too big.

What this demonstrates

- Cross-referencing records (sampling-point -> samplingPointType).
- Identifying inaccuracies, inconsistencies, and missing data at scale.
- Root-cause attribution (methodology accretion over decades) rather than reflexive cleaning.
- Caveats alongside findings so non-technical readers understand what the numbers do and do not mean.
- A Power BI-equivalent one-page visual summary for stakeholders.

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Source: github.com/Leonw98/ea_wqa_audit (MIT licence).
Reusable data-quality work sample, April 2026.

